

ALUVA STATE SEED FARM
Kochi

ASHISH K VINCENT

"We have followed organic modules on the farm since 2012, while also sensitising farmers about reducing greenhouse gas emissions"

Lissymol J. Vadakkoot, *Asst. director, Kerala agriculture dept.*

Experts from the Kerala Agricultural University are conducting soil sampling and other tests now. The agriculture ministry says the 13 seed farms will receive carbon-neutral certification by June 2024. "We are planning to set up carbon-neutral farms in all 140 assembly constituencies to encourage organic farming. By following the integrated farming model, the costs will also come down," says agriculture minister P. Prasad, a farmer himself.

The integrated farming system is defined as a self-sustaining agricultural model that integrates livestock, crop production, fish, poultry, trees, plantation crops and other systems so that they are mutually beneficial. The farm animals and crops are reared in a mix-and-match fashion so that they complement each other and increase the farm's productivity.

For instance, duck farming and pisciculture were found to give a huge boost to paddy cultivation. 'Rice productivity showed a tremendous rise when the farm started rearing ducks and certain kinds of fish. In fact, the ducks saved 100 labour days and improved the fertility of the land,' says an assessment by the expert team. The ducks and fish produced great manure and also minimised the problem of pests like stem borers, leaf rollers and plant hoppers in the paddy fields.

Meanwhile, in the dryland areas, the desi cows come into play, providing slurry for the horticulture crops while the vegetable waste provides fodder. Apart from the native cows, Malabari goats, apiaries and poultry also play a key role on the farm. The farm uses organic pesticides brewed form local materials while the stubble and waste are turned into vermi-compost. Even the seeds used for propagation are produced according to "ecological engineering methods". "We use scientific methods to counter the greenhouse effect. The intensity of carbon emission depends on the organic substance content in the soil, the rate of mineralisation, the presence of microorganisms and the moisture level," Vadakkoot points out.

The seed farm's ability to showcase innovative farming techniques, including improved crop habitat and yield, natural pest control, crop rotation and profitable animal husbandry, has made it popular among farmers and owners with small land holdings looking to increase their incomes. The centre also has a live rice museum with different varieties being grown such as the medicinal red rice Rakthashali, nutrient-rich Japan violet, the traditional varieties of Vellathondi, Njavaras and others. Chief Minister Pinarayi Vijayan says he wants Kerala to be a net carbon-neutral state by 2050. It's an ambitious project but a small beginning has been made with the carbon-neutral farms. ■